

Using Visual Studio 11
and Scrum to plan
and track ~~a Project~~
software product
development



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The Professional Scrum Developer Program



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Scrum on a Slide

► Roles

- Product Owner, Scrum Master, Development Team

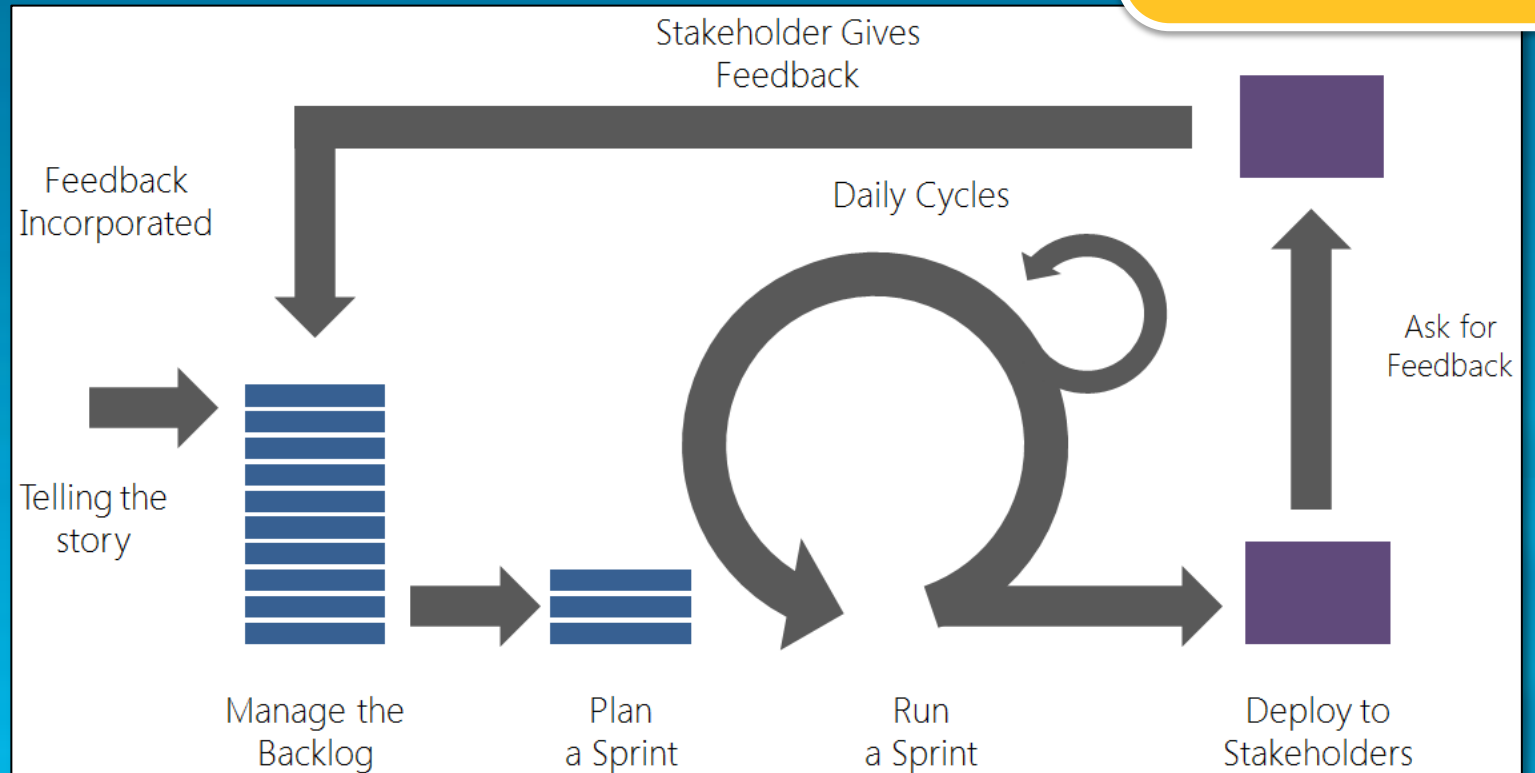
► Events

- The Sprint
- Sprint Planning
- Daily Scrum
- Sprint Review
- Sprint Retrospective

► Artifacts

- Product Backlog
- Sprint Backlog
- The Increment of Working Software

According to
July 2011
Scrum Guide



Visual Studio Scrum 2.0 Artifacts

Work Item Types	Queries	Reports
Product Backlog Item	Product Backlog	Backlog Overview
Task	Current Sprint	Release Burndown
Code Review Request	Blocked Tasks	Sprint Burndown
Code Review Response	Open Impediments	Velocity
Feedback	Sprint Backlog	Build Success Over Time
Impediment	Test Cases	Build Summary
Test Case	Unfinished Work	Test Case Readiness
Shared Steps	Work in Progress	Test Plan Progress
Sprint	My Work	
	...	

Visual Studio Scrum Activities

- ▶ Scrum Team Projects have many standard and repetitive activities
 - ▶ Setting up a new Team Project Collection
 - ▶ Setting up a new Team Project
 - ▶ Setting up a new Sprint
 - ▶ Daily activities
 - ▶ Completing a Sprint
- ▶ Note: Some activities will be done in Windows (TFSSAC, TE) and the Web (the new TWA)

Activities: Setting up a Team Project Collection

- ▶ Create the new TPC (if necessary)
- ▶ Configure SharePoint, reporting, Lab, etc.
- ▶ Configure Team Foundation Build for that collection
- ▶ Configure security membership and permissions
- ▶ Upload Visual Studio Scrum template

Activities: Setting up a ~~Team Project~~ Product

- ▶ Create the Team Project using the Scrum process template
- ▶ Configure security groups, permissions, team, photos
- ▶ Configure SharePoint, process guidance, and version control settings
- ▶ Configure the work item areas and iterations
- ▶ Record the Definition of Done in SharePoint (document or wiki)
- ▶ Setup version control folder structure and add files
- ▶ Migrate/create work items to create the initial Product Backlog
- ▶ Use Team Web Access

Speaking of the Product Backlog ...

Specifying Acceptance Criteria

- ▶ This can be done incrementally

Disambiguity

1. Initially, a PBI can just have a title
2. Later, the PBI can be updated to include a more detailed description
3. Later, the PBI can be updated to include detailed acceptance criteria
4. Later, the PBI can be linked to a Test Case work item with more detailed criteria in its description field
5. Later, the Test Case can have steps specified (MTM) and Manual tests can be recorded and automatically played-back (MTM)

Estimating Effort

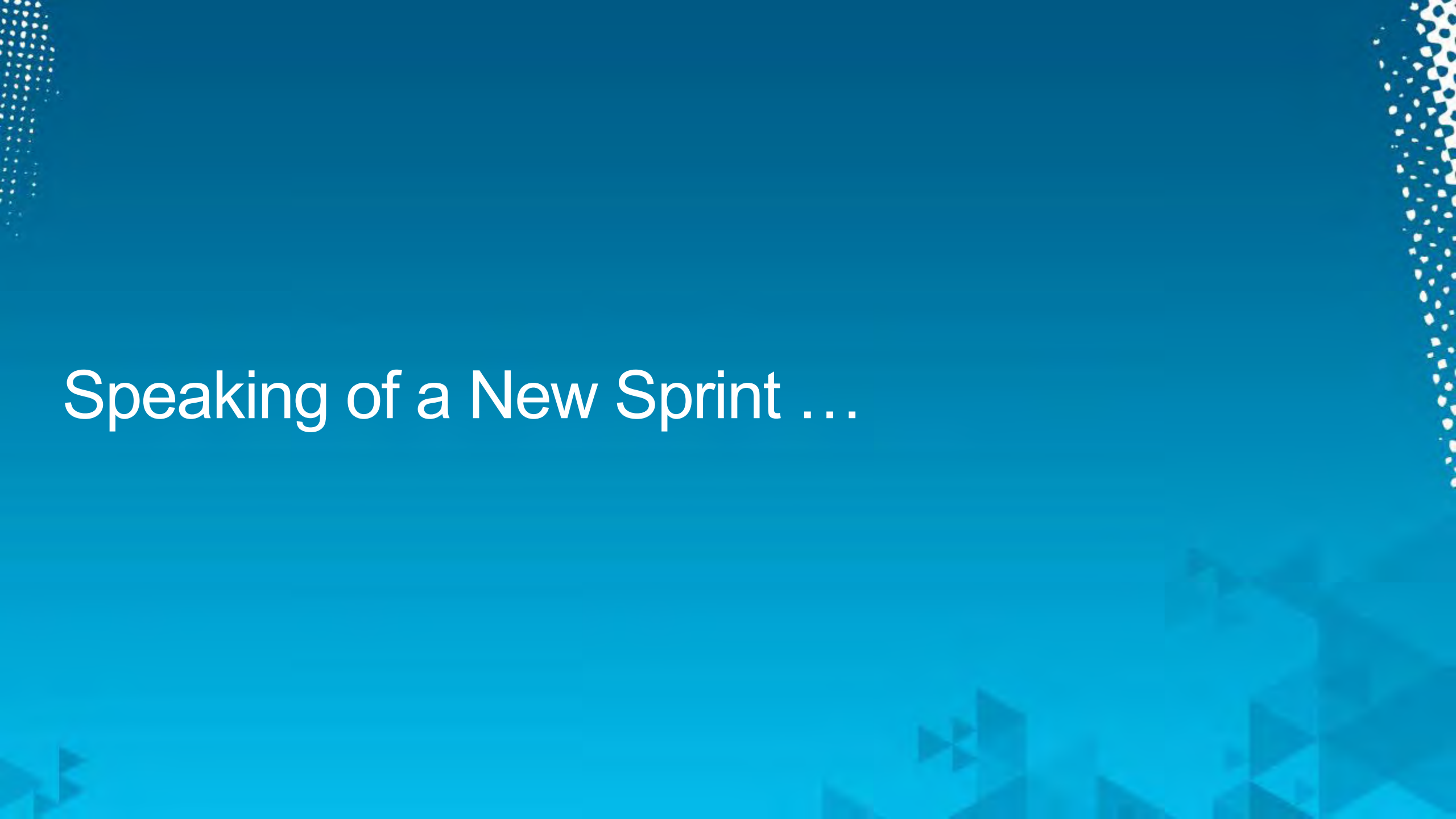
- ▶ The Team must estimate the level of effort it will take to implement each item in the Product Backlog
 - ▶ Estimation should be done collaboratively
 - ▶ Estimates are not a plan or a commitment, just an estimate
- ▶ Story Points
 - ▶ A unit of measure for expressing the overall size of a PBI
 - ▶ Points are relative and do not translate into precise hours
 - ▶ Only Sprint tasks are estimated in hours
 - ▶ Planning Poker® is a collaborative estimation technique

Grooming the Product Backlog

- ▶ Anyone can contribute to the Product Backlog
 - ▶ The Product Owner provides business value
 - ▶ The team can provide costs and dependency details
 - ▶ The Product Owner is responsible for prioritization
- ▶ Grooming the Product Backlog
 - ▶ You want to have prioritized PBIs ready for the next Sprint
 - ▶ You may even need to pull new PBIs into the current Sprint
 - ▶ Keep the Product Backlog in good shape at all times
 - ▶ When? Let the team decide, but make sure they do and devote $\leq 10\%$ of the Sprint length

Activities: Setting up a New Sprint

- ▶ Ensure that the release and Sprint nodes exist
- ▶ Ensure that the node has start/end dates
- ▶ Modify the Current Sprint queries
- ▶ Run the Product Backlog query for accuracy
- ▶ Review the Release Burndown and Velocity reports
- ▶ Edit the PBI work items you are committing to deliver
- ▶ Create and link initial task work items
- ▶ Review the Sprint Backlog query for accuracy



Speaking of a New Sprint ...

Sprint Planning

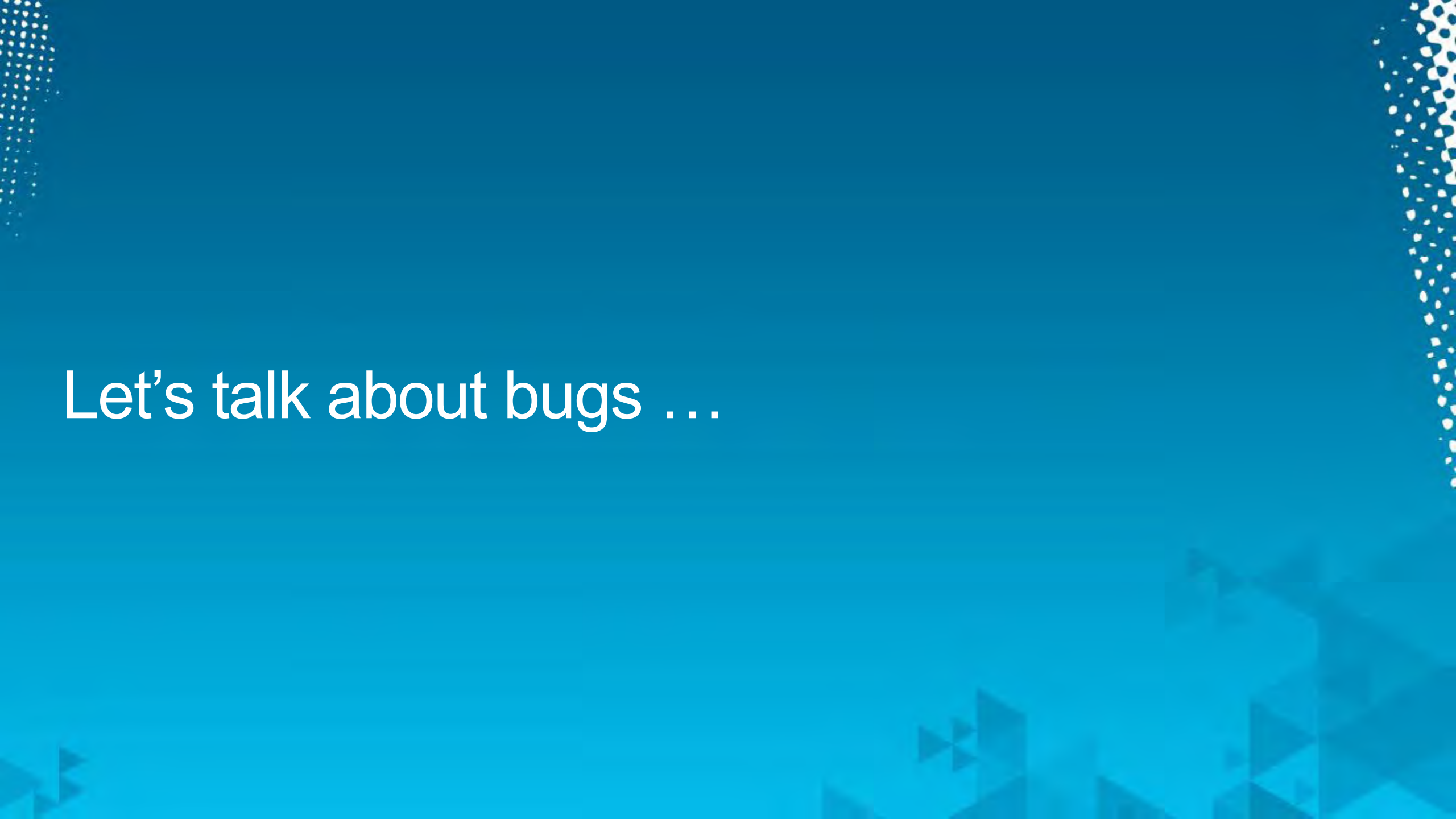
- ▶ Determine the work that can be completed during the Sprint
 - ▶ Product Backlog must be in good shape prior to this
- ▶ Two parts to planning:
 1. Choose a Goal – the Scrum Team collaborates to forecast how much of the ordered (prioritized) backlog can be turned into potentially shippable functionality
 2. Create Sprint Backlog – the team defines the plan (tasks) required to build that functionality during the Sprint, including estimates to achieve the Definition of Done

Sprint Backlog

- ▶ The Sprint Backlog contains the PBIs/Bugs and the related tasks that define the team's work for a Sprint
 - ▶ Tasks emerge during Sprint planning
- ▶ The Sprint Backlog is what the Team must complete to meet the Sprint Goal
- ▶ The Sprint Burndown chart is based on the Sprint Backlog being accurate and up-to-date
 - ▶ Estimate work remaining (hours) on your tasks daily

Activities: Daily Scrum Tasks

- ▶ Create and link task work items as needed
- ▶ Team takes on (signs up for) tasks from the Sprint Backlog
- ▶ Check-in any code changes to TFS associating with one task only
- ▶ Update remaining work estimates on your tasks
- ▶ Complete your work according to the acceptance criteria and DoD
- ▶ Set the state and remaining work on completed tasks
 - ▶ The (task) board is a great tool for this
- ▶ Update the Product Backlog by creating and editing PBI and Bug work items as needed (grooming)
- ▶ Create and manage impediment work items as needed
- ▶ Review the Sprint Burndown report to monitor progress

The background is a solid blue color. In the top-left and top-right corners, there are decorative halftone patterns consisting of small white dots. At the bottom, there is a faint, low-poly geometric pattern in a lighter shade of blue.

Let's talk about bugs ...

What Makes a Good Bug Report

- ▶ Clear title
- ▶ One bug per report
- ▶ Simple and repeatable reproduction steps
- ▶ Expected results and observed results
- ▶ Build/version/environment (Dev, QA, UA, Prod) where bug was found
- ▶ Identify the product area/component
- ▶ Link to other related bugs (reactivation?)
- ▶ Attached screenshots and other artifacts
- ▶ Proper grammar, spelling, tone – no politics

How to Fix a Bug

1. Verify the bug by writing a failing test
2. Fix the bug by making the test pass
3. Verify the bug is fixed by re-running all tests
4. Refactor according to your definition of Done
 - ▶ Avoid “gold-plating”
5. Re-run all tests to ensure no other problems
6. Merge your code with the latest version in TFS, build, and re-run all tests
 - ▶ Team Foundation Build can automate this
7. Check-in code changes and tests
 - ▶ Associate with the “fix the bug” Task work item
8. Close the Task and Bug work items

Activities: Completing a Sprint

- ▶ Label code and other artifacts in version control
- ▶ Set the State of the PBI and/or Bug work items to Done when the Product Owner accepts it
- ▶ Set the State of the PBI/Bug and linked work items appropriately for unfinished/rejected work
- ▶ Update the Product Backlog by creating and editing PBI and Bug work items as needed
- ▶ Capture the Sprint Retrospective feedback on the Sprint work item
- ▶ Branch as needed (i.e. to support hotfixing)

Let's be professionals ...

"Done"

- ▶ Done defines when an increment of product functionality is potentially shippable
- ▶ Definition of Done (DoD)
 - ▶ A simple, auditable checklist owned by the team
 - ▶ It can be influenced by organizational standards and specific requirements of the product or release
- ▶ Here is a simple Definition of Done
 1. Code complete
 2. Tests pass
 3. Everything checked-in
 4. Acceptance criteria from Product Owner met

Don't be *Flaccid*

- ▶ Martin Fowler describes Flaccid Scrum (<http://bit.ly/17aEU>)
- ▶ Flaccid teams believe in magic
 - ▶ They don't have skills and supportive infrastructure
 - ▶ They are using inadequate practices and tooling
 - ▶ They hide behind the waterfall
- ▶ Flaccid customers believe in magic
 - ▶ They throw a big book of requirements over the wall
 - ▶ They avoid inevitable and unavoidable changes
 - ▶ They think that pressuring the team will work
- ▶ Don't just inspect ... adapt!

Done



Learn more here: www.scrum.org

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